

FlowSync: Traffic Simulation System Software

Design Report

Project Team

Abstract

FlowSync is a modular traffic simulation system designed using Object-Oriented Programming principles. The system models vehicle movement, lane management, and traffic signals while ensuring scalability, maintainability, and clean architecture. The implementation follows SOLID principles and incorporates design patterns such as Strategy, Observer, Factory, and Template Method. The system is fully integrated, tested, and capable of real-time simulation and visualization.

1 Introduction

Traffic simulation plays an important role in understanding real-world traffic behavior, optimizing signal timing, and improving road efficiency. FlowSync is developed as a modular simulation system that replicates real-world traffic flow using vehicles, lanes, and traffic signals.

The primary goal of this project is to design a system that is:

- Scalable
- Maintainable
- Extensible
- Architecturally clean

2 Problem Statement

Traditional traffic simulation systems often suffer from tightly coupled components, making them difficult to extend and maintain. The problem addressed in this project is to design a traffic simulation system that:

- Separates responsibilities clearly
- Allows easy integration of new features
- Maintains stable behavior under load

3 Objectives

- Design a modular traffic simulation system
- Implement vehicle behavior using physics models
- Simulate traffic signals and intersections
- Ensure system stability and correctness
- Apply SOLID principles and design patterns

4 System Architecture

4.1 Overview

SimulationController



TrafficManager



Road



Lane



Strategy



Vehicle



Physics + Braking

Intersection → TrafficSignal → Vehicle

4.2 Description

The system follows a layered architecture where each component has a specific role. The TrafficManager orchestrates updates, while Lane manages vehicle interactions and Vehicle handles physics and behavior.

5 Module Description

5.1 Traffic Manager

- Controls update loop
- Ensures signals update before vehicles
- Coordinates all system components

5.2 Vehicle System

- Handles motion and physics
- Implements IDM model
- Applies braking logic
- Responds to traffic signals

5.3 Lane System

- Maintains ordered list of vehicles
- Detects lead vehicle
- Delegates update using Strategy pattern
- Ensures deterministic updates

5.4 Traffic Signal

- Implements signal state transitions
- Uses Observer pattern to notify vehicles

5.5 Visualization / Scene

- Renders vehicles and roads
- Provides camera control
- Enables interactive simulation

6 SOLID Principles

6.1 Single Responsibility Principle

Each component has a single responsibility:

- Vehicle handles physics
- Lane handles ordering
- Signal handles state

6.2 Open/Closed Principle

System supports extension without modification:

- New strategies
- New physics models

6.3 Liskov Substitution Principle

Lane interacts with abstract vehicle interface.

6.4 Interface Segregation Principle

Vehicle interface is minimal and focused.

6.5 Dependency Inversion Principle

Lane depends on Strategy abstraction rather than concrete implementations.

7 Design Patterns

7.1 Strategy Pattern

Lane delegates update logic:

Lane → Strategy → Vehicle

7.2 Template Method

```
update():  
    pre_update()  
    process()  
    post_update()
```

7.3 Observer Pattern

Signal → notify → Vehicle

7.4 Factory Pattern

Lane creation is handled by factory.

8 Data Flow

```
TrafficManager.update()  
  → Intersection.update()  
  → Signal.update()  
  → Signal.notify()  
  → Road.update()  
  → Lane.update()  
  → Vehicle.update()
```

9 Testing and Validation

- Unit tests for vehicle behavior
- Integration tests for lane interaction
- Stress tests with multiple vehicles
- Validation checks:
 - No vehicle overlap
 - Sorted lane order

10 Results

- Stable simulation achieved
- No collisions under stress conditions
- Vehicles respond correctly to signals
- System performs efficiently

11 Future Work

- Lane changing behavior
- Advanced AI-based traffic control
- Real-time data integration
- Improved visualization

12 Conclusion

FlowSync successfully demonstrates a clean and modular traffic simulation system. The application of SOLID principles and design patterns ensures maintainability and extensibility. The system is stable, efficient, and suitable for demonstration and further development.